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Lower–Middle Jurassic paleomagnetic data from the Mae Sot area (Thailand): Paleogeographic evolution and deformation history of Southeastern Asia

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Abstract

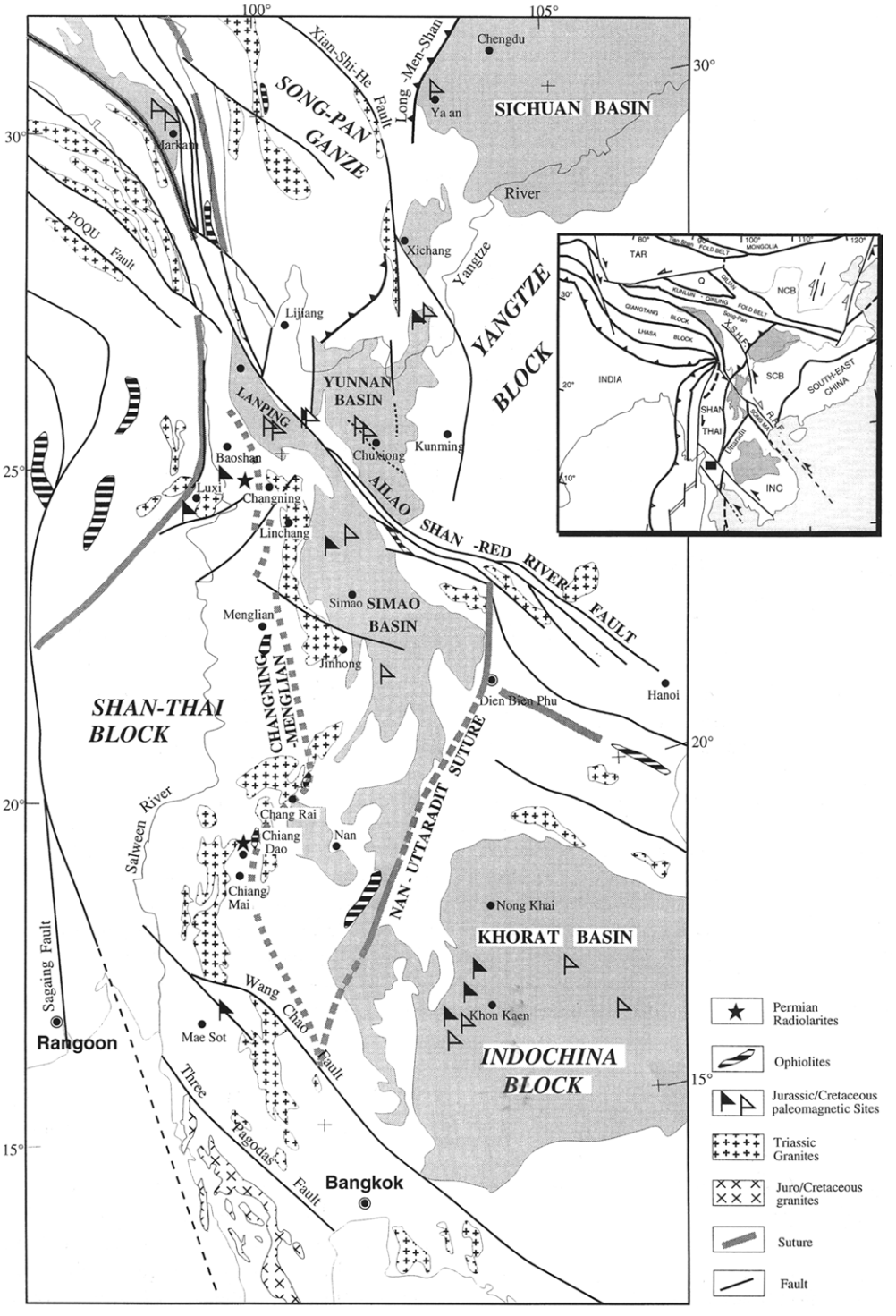
We have carried out a paleomagnetic study (12 sites, 85 samples) of Early–Middle Jurassic limestones and sandstones from the Mae Sot area of western Thailand. This area is part of the Shan–Thai–Malay (STM) block, and its geological characteristics have led some authors to suggest a Late Jurassic accretion of this region against the rest of Indochina along the Changning–Menglian zone, the latter sometimes being interpreted as a Mesozoic suture. The high-temperature (or high-coercivity) component isolated yields a paleodirection at $D = 359.8°$, $I = 31.4°$ ($\alpha_{95} = 5.0°$). The primary nature of the magnetization acquisition is ascertained at a site with reversed polarity and a positive fold test (at the 95% confidence level). Comparison of the Mae Sot paleolatitude and another one from the STM with those recently published for the Simao and Khorat blocks show no significant difference at the 95% level, showing that the STM was situated close to, or had already accreted with, the Simao or Khorat blocks in the Early–Middle Jurassic. Comparison of the latitudes from these blocks with those from China indicates a relative southward motion of $8 \pm 4°$ of Indochina as a single entity relative to China. Most rotations of these regions relative to China are found to be clockwise (between 14 and 75°). These rotations, and most prominently the 1200 ± 500 km post-Cretaceous left-lateral motion inferred for the Red River Fault, provide quantitative estimates of the large amount of extrusion of Indochina with respect to the rest of Asia.

1. Introduction

Indochina is a mosaic of blocks which had accreted by the end of the Paleozoic and during the Mesozoic [1,2]. The number of terranes and the timing of their collision are still debated. The Shan–Thai–Malay (STM) block (Fig. 1), which is of Gondwanian origin [1,2], is separated from the Khorat block by the Nan–Uttaradit suture, whose prolongation farther south is the Raub–Bentong suture of Peninsular Malaysia. The age of this suture, which is mainly based on the age of S-type collision-related

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100°
105°
30°
25°
20°
15°
Xian-Shi-He Fault
Long-Men-Shan
Chengdu
SONG-PAN GANZE
Ya'an
SICHUAN BASIN
Markam
River
POQU Fault
Xichang
Yangtze
YANGTZE BLOCK
Lijiang
YUNNAN BASIN
LANPING
Chuxiong
Kunming
Baoshan
Luxi
Changning
AILAO SHAN-RED RIVER FAULT
Linchang
Simao
SIMAO BASIN
Menglian
Jinhong
CHANGNING-MENGLIAN
SHAN-THAI BLOCK
Dien Bien Phu
Hanoi
Salween River
Chang Rai
Chiang Dao
Nan
NAN-UTTARADIT SUTURE
Chiang Mai
Nong Khai
KHORAT BASIN
Sagaing Fault
Wang Chao Fault
Khon Kaen
Rangoon
Mae Sot
INDOCHINA BLOCK
Three Pagodas Fault
Bangkok
TAR
NCB
SCB
SOUTHEAST CHINA
INDIA
SHAN THAI
INC
Permian Radiolarites
Ophiolites
Jurassic/Cretaceous paleomagnetic Sites
Triassic Granites
Juro/Cretaceous granites
Suture
Fault

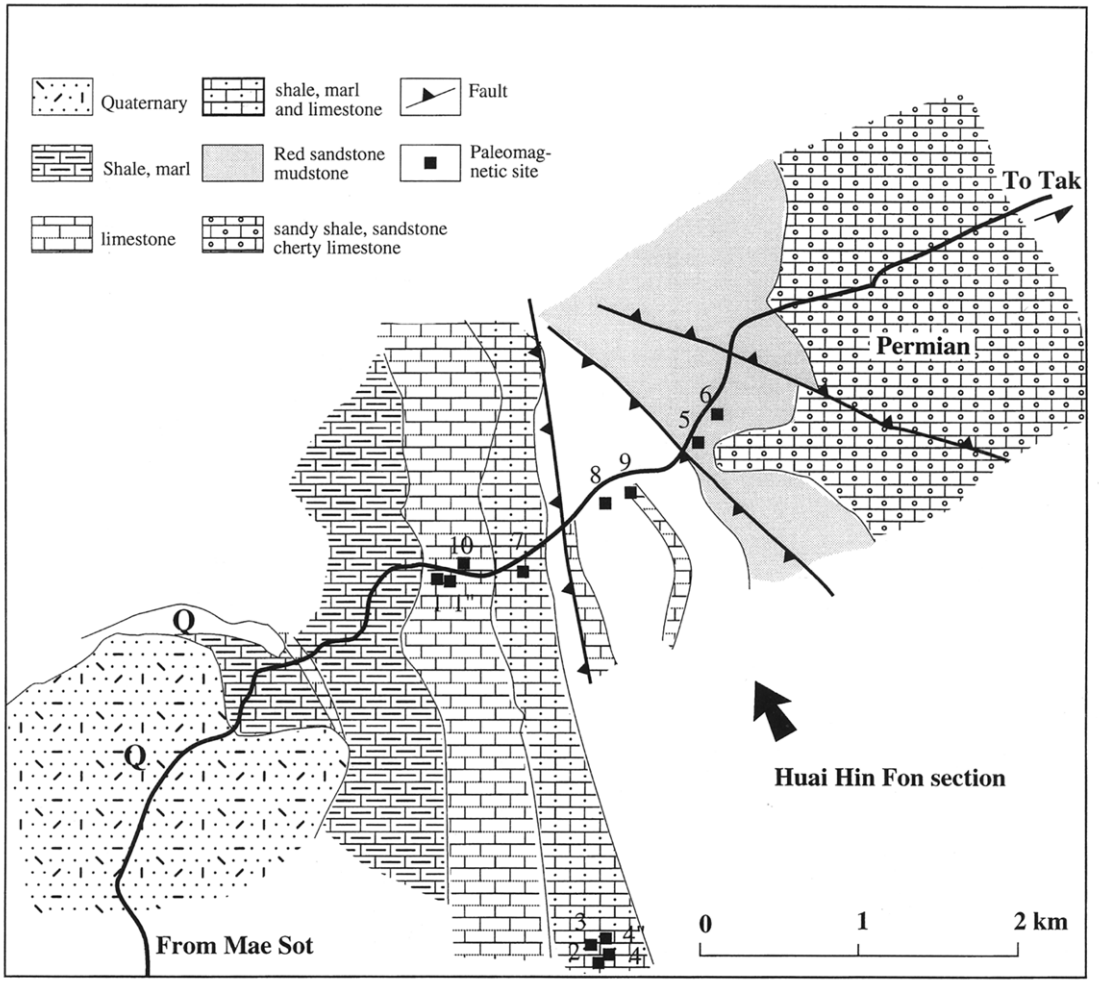


Fig. 2. Simplified geological map after Bunopas [22] of the Mae Sot basin, western Thailand (Shan-Thai block), with sampling sites indicated (■).

granites situated west of Nan Uttaradit and in the western part of peninsular Malaysia, is probably Late Triassic [2–5]. This suture possibly continues to the northeast, following a direction roughly parallel to the Dien Bien Phu right-lateral strike-slip fault. It then passes to the southeast of the Simao basin and joins farther east with the Ailao Shan region [6,7]. The STM block is delimited to the west by the Shan boundary fault and to the southwest probably by the Sumatran fault.

A connection of the southern part of the Nan Uttaradit suture with the Changning–Menglian zone has also been suggested [8–12]. Recent geological surveys have identified sediments of pelagic character of Devonian–Middle Triassic age at several localities in South Yunnan (Changning–Menglian) and in

Fig. 1. Tectonic sketch map of Southeast Asia. Three main blocks (Shan–Thai, Simao and Khorat) are recognized. Inset: Sketch map of the tectonic elements of eastern Asia, modified after Yang et al. [16] and Peltzer and Tapponnier [6]. Major blocks: *TAR* = Tarim; *NCB* = North China Block; *SCB* = South China Block; *Q* = Qaidam; *JUN* = Junggar; *IND* = India; *INC* = Indochina block. Arrows correspond to the sense of strike-slip motion on major faults (open for recent, closed for older phases).

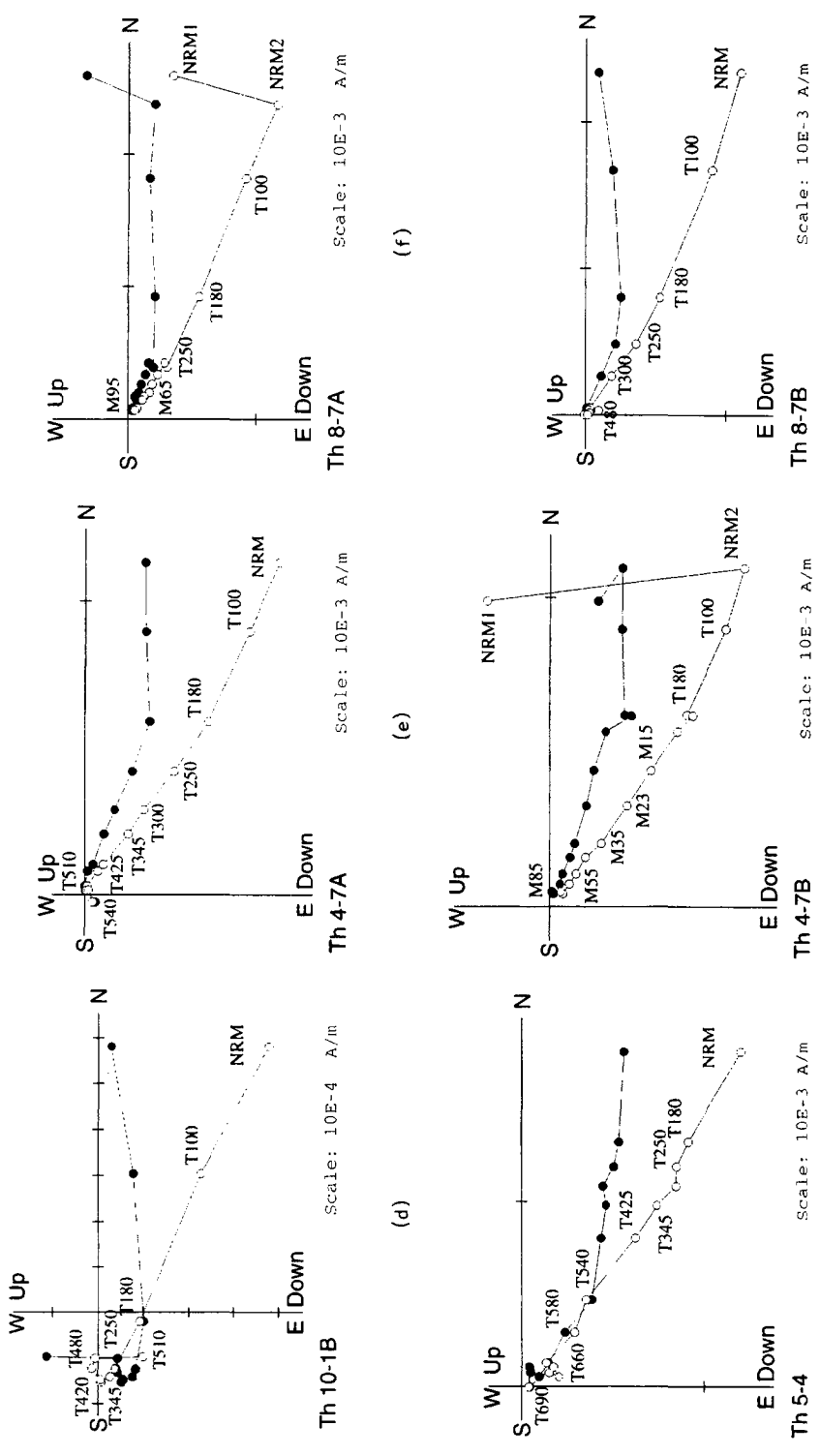


Fig. 3. Orthogonal vector projection for thermal (a,b,d,f) and mixed (c,e) demagnetization data on Lower and Middle Jurassic samples from the Mae Sot basin. Solid/open symbols refer to projection on the horizontal/vertical plane in geographic coordinates.

northwest Thailand (Chiang Dao area) that may indicate an important N–S suture [11] (Fig. 1). This suture is a barrier which separates a zone of Permo-Carboniferous Gondwanian affinities to the west from a zone of Cathaysian affinity to the east. The STM would thus comprise two blocks, the Shan–Thai–Malay *sensu stricto* on the western side of the Changning–Menglian line, and an intermediate block, the Simao block, bounded to the southeast by the Nan–Utaradit suture. The collision of the STM block against eastern Indochina would thus be more complex than previously thought: a collision of the intermediate Simao block against the Khorat zone would have occurred sometime during the Late Triassic, while a second collision of the western STM would occur in a poorly defined period ranging from the Late Triassic [10] to the Late Jurassic [11–13]. This later age is based on a clear distinction of facies on either side of this suture—continental sediments are found on the eastern side while marine sediments are found on the western side. In particular, marine faunas of middle–late Toarcian age that are similar to those of South Tibet are found in the Mae Sot and Kanchanaburi provinces of Thailand, leading to the idea that the western part of the Shan–Thai is the eastern continuation of South Tibet. This late collision would be coeval with the accretion, beginning in the Late Jurassic, of the Lhasa block with North Tibet along the Bangong Nujiang suture zone [14,15].

Paleomagnetism has proven to be an independent means of constraining the paleoposition and tectonic evolution of crustal blocks. In this paper, we present new paleomagnetic results from the study of marine sediments of Lower-early Middle Jurassic age from the Mae Sot area, which is situated west of the Changning–Menglian zone in western Thailand. The aim is to track possible latitudinal motions with respect to the Simao block and the Khorat region from where paleomagnetic data have recently been published [16–18].

Models of continental deformation after the onset of the collision are a matter for much debate [6,7,19]. A previous paleomagnetic study of Mesozoic sediments from the Khorat plateau (Thailand) supported the extrusion tectonics model proposed by Tapponnier et al. [16]. Our new data, together with additional data from Indochina and China, allow further comparison of the paleolatitudes and rotations of blocks situated on either side of the Red River Fault (RRF), and will help to refine the amount and timing of post-Cretaceous left-lateral motion on this fault.

2. Geological setting

We have sampled uppermost Triassic to lower Middle Jurassic sediments in the Mae Sot basin along the Mae Sot–Tak highway close to the Thailand–Burma border (Fig. 2). This section has previously been studied by Von Braun and Jordan [20] and Fontaine and Suteethorn [21]. The succession is gently folded (30° along an approximately north–south trend) and is discomformably overlain by a Mesozoic red conglomerate (of probable Cretaceous age) in the southeastern part of the Mae Sot basin [22]. The uppermost Triassic–Lower Jurassic sediments (Mae Moei Group) unconformably overlie the fossiliferous Lower Permian limestones and are represented by fine-grained terrigenous clastic sediments interbedded with marine limestones. Twelve

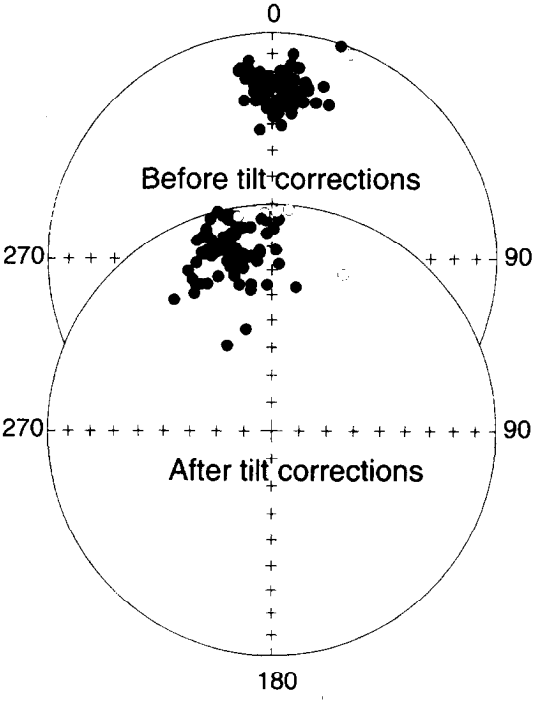


Fig. 4. Equal-area projection of the recent overprint direction before and after tectonic tilt correction.

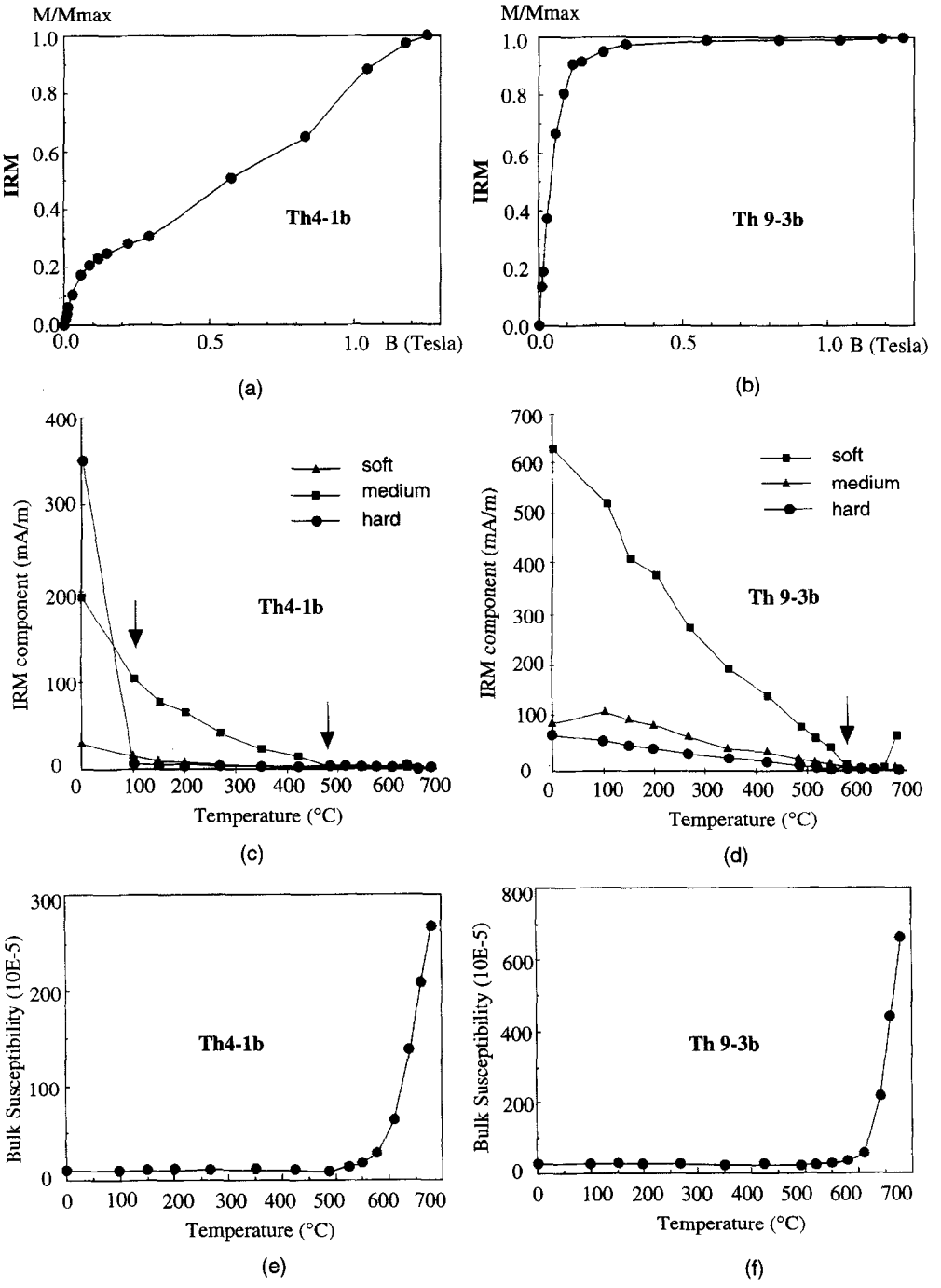


Fig. 5. (a,b) Normalized isothermal remanent magnetization (IRM) acquisition curves. (c,d) Three-component IRM thermal demagnetization showing unblocking temperatures around 100°C and 510–580°C. (e,f) Change in the bulk susceptibility during heating for the same specimens.

sites have been sampled along the section. Sites 5 and 6 consist of grey sandstones close to the Triassic–Jurassic boundary, as determined from palynological data [21]. Site 7 consists of grey limestones with a wackestone microfacies. The age (early or middle Toarcian) of this rock has been assigned on the basis of its stratigraphic position below Site 1. At Sites 8 and 9, grey-black micritic limestones outcrop. The age of these limestones is not precisely known because of the lack of good stratigraphic fossils, but their stratigraphic position below middle Toarcian beds and the presence of the alga *Holosporella siamensis* suggests a probable early Toarcian age. Sites 1 and 10 consist of black limestones with a wackestone microfacies. A middle Toarcian age is assigned on the basis of fragments of ammonites (*Phymatoceras* sp., Hammatoceratinae). Sites 2 and 3 are grainstone/packstone limestones. These are dated to the Aalenian/Bajocian on the basis of the presence of the foraminiferae *Timidonella* cf. *sarda*, *Spiraloconulus* sp. and *Bosniella ontainei* [13]. Finally, well-bedded limestones are found at Site 4, and again the presence of *Timidonella* indicates an Aalenian to Bajocian age.

A total of 75 separate cores (85 laboratory samples) from twelve sites (ten sites and two subsites) have been collected using a portable petrol-powered drill and oriented using a magnetic compass.

3. Paleomagnetic study

Natural remanent magnetization (NRM) was measured using a CTF cryogenic magnetometer in the shielded room at the IPG in Paris. Stepwise thermal and/or alternating field (AF) demagnetizations were carried out up to 690°C (or 100 mT) with a laboratory-built oven or a Schonstedt GSD1 single-axis demagnetizer. A mixed thermal–AF demagnetization was sometimes used for some rocks containing magnetite as a remanence carrier. The bulk magnetic susceptibility of the specimens was measured at each demagnetization step in order to monitor magnetic mineral transformations during the progressive heating. The magnetic directions were analyzed by principal component analysis [23], and the site-mean direction was calculated using Fisherian statistics [24].

The orthogonal demagnetization diagrams show relatively simple magnetic behaviour. Two magnetic components may be isolated (Fig. 3). A first component loses magnetization above 200–250°C or in a field above 30 mT. The mean direction of this component ($D = 1.1°$, $I = 24.8°$, $\alpha_{95} = 3.1°$, $N = 12$ sites), which shows a negative fold test at the 95%

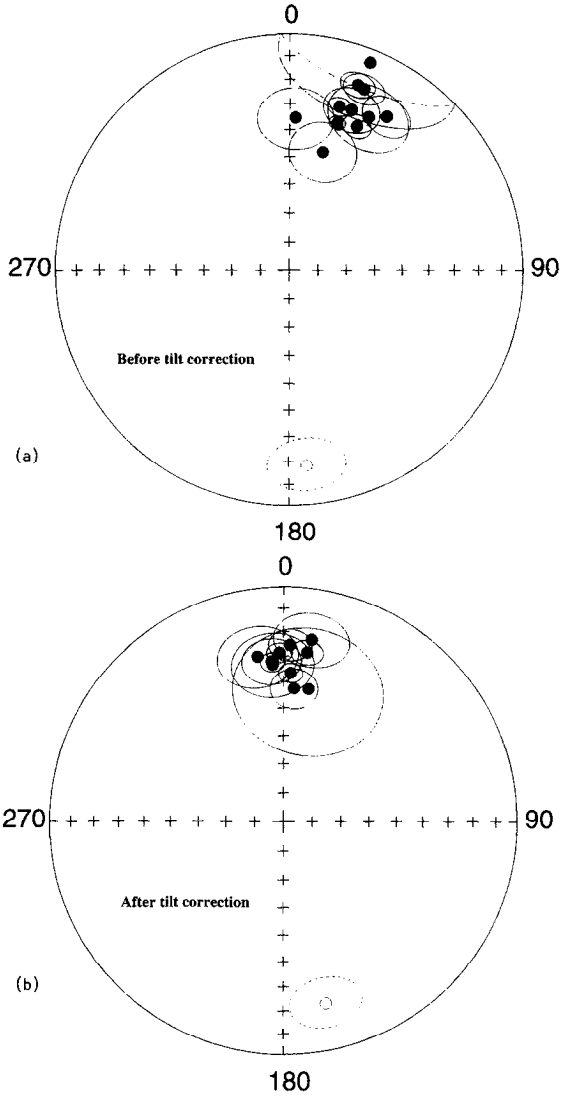


Fig. 6. Equal-area projection of the Lower Jurassic site-mean characteristic remanent magnetization (ChRM) directions of the Mae Sot basin before (a) and after (b) tectonic correction. Solid/open symbols represent directions plotted onto the lower/upper hemiphere. Circles are 95% confidence limits around the site-mean direction.

confidence level ($\kappa_g/\kappa_s = 5.3$), is situated between the present local dipole field ($D = 0°$, $I = 31°$) and the present field computed using the IGRF ($D = 359°$, $I = 19°$) and probably corresponds to an overprint (Fig. 4). The higher temperature/coercivity component gently decreases toward the origin until 500–580°C, or around 80 mT, for eleven sites (Fig. 3a–d). A mixed demagnetization (thermal demagnetization until 180°C, then by AF up to 85 mT) was also carried out on some samples (Fig. 3e) and resulted in the characterization of a final component identical to that obtained by thermal demagnetization only (the two techniques on two samples from the same core are compared in Fig. 3d and e). Site 5 is composed of red siltstone and Fig. 3f shows that the magnetization is lost at around 690°C.

In order to better characterize the magnetic fraction, we have studied the acquisition of isothermal remanent magnetization (IRM) and its thermal demagnetization [25] for several specimens from sites of various petrologies. The first (Fig. 5a, Site Th4) shows that despite a first saturation step around 0.20–0.30 T, a field of 1.26 T is not sufficient to reach complete saturation. In contrast, Fig. 5b (Site Th9) shows that saturation is reached before a fairly low field of 0.3 T. A different DC field (1.26, 0.4 and 0.12 T) is applied successively to each of three perpendicular axes of a sample and then thermally demagnetized. Two main types of behaviour can be seen. The results are shown in Fig. 5c and d. A large drop in intensity at 100°C on the high-coercivity component suggests the presence of goethite in the sample of Site 4 and not in the sample of Site 9. The magnetization is then dominated by a medium–low-coercivity mineral (soft coercivity is predominant) and the intensity gently decreases to 500–580°C. The susceptibility measurement (Fig. 5e,f) indicates no noticeable transformation of the magnetic fraction during thermal demagnetization to 550–580°C. These characteristics suggest that a mineral of the titanomagnetite family is the carrier of the final component.

The red siltstone site (site 5, Fig. 3f) shows a higher unblocking temperature above 650°C, which suggest that hematite is the principal magnetic carrier. Similar characteristic directions are found in

Table 1
Characteristic remanent magnetization directions for the Mae Sot sites (sampling location at 16.75°N, 98.67°E)

Site	rocks unit	Strike/Dip	N	Dg	Ig	Ds	Is	κ	α95
Th1	Limestone	185°/48°	8	22.5	17.9	2.1	25.5	76.1	6.4
Th1"	Limestone	165°/50°	5	20.3	17.2	358.5	28.6	376.7	3.9
Th2	Limestone	175°/44°	7	32.5	23.7	4.5	42.6	64.3	7.6
Th3	Sandstone	180°/40°	9	21.3	27.7	355.6	31.9	34.8	9.1
Th4	Limestone	180°/14°	5	2.2	35.4	350.9	29.4	39.6	12.3
Th4"	Limestone	180°/22°	11	18.0	34.5	2.7	36.7	149.0	3.8
Th5	Sandstone	139°/42°	4	21.4	6.2	10.7	42.1	15.2	24.3
Th6	Limestone	260°/26°	5	15.8	46.6	8.8	22.5	55.6	11.1
Th7	Limestone	186°/50°	4	27.6	27.6	355.9	33.2	53.3	12.7
Th8	Limestone	218°/28°	13	25.2	32.9	7.9	28.0	71.5	4.9
Th9	Limestone	180°/35°	5	17.0	28.1	356.0	32.3	440.4	3.8
Th10	Limestone	168°/23°	9	174.7	-18.0	166.7	-21.2	24.5	10.6
Mean of normal sites			11	20.6	27.3	1.1	32.2	94.2	4.7
Mean of all sites			12	18.4	26.7	359.8	31.4	75.7	5.0

D/I = declination/inclination ("g" = geographic, "s" stratigraphic). $_s/\kappa_g = 2.3 > F(22,22) = 2.05$ indicates a positive fold test at the 95% confidence level

both magnetite and hematite carriers (Fig. 3). Out of twelve sites, eleven are of normal polarity and one is of reversed polarity (Fig. 6). The direction of the reversed site is nearly anti-parallel to the mean direction of the other eleven normal sites (Fig. 6). The site-mean ChRM directions of the twelve sites yield a north downward direction, both in stratigraphic coordinates ($D_s = 359.8°$, $I_s = 31.4°$, $\alpha_{95} = 5.0°$) and in field coordinates ($D_g = 18.4°$, $I_g = 26.7°$ ($\alpha_{95} = 7.7°$) (Fig. 6, Table 1). The progressive increase in the precision parameter (κ) during unfolding ($\kappa_s/\kappa_g = 75.7/32.9$ at 100% unfolding) leads to a positive fold test at the 95% confidence level [25] and suggests that the ChRM direction pre-dates the folding (Table 1).

4. Discussion

We have successfully isolated a high-temperature component from the Lower to Middle Jurassic sediments of the Mae Sot area. In both lithologies (sandstone and limestone), this component appears to be carried by a mineral of the titanomagnetite family (with the exception of Site 5 for which the magnetic carrier is hematite). Although the corresponding virtual geomagnetic pole is located close to the geographic north pole (Fig. 6), the reversed site and the positive fold test at the 95% level suggest that the higher blocking component is of primary origin, or at least of pre-Cretaceous age, since Lower Cretaceous sediments are discordant over the Jurassic [22]. A paleolatitude of 17°N ± 3° can be derived and the paleodeclination ($D = 0°$), indicating no rotation relative to the earth's axis of spin. We may now compare our directions with those already available in the region.

4.1. Comparison with other paleomagnetic data from Indochina and the Simao and Shan–Thai–Malay (STM) blocks

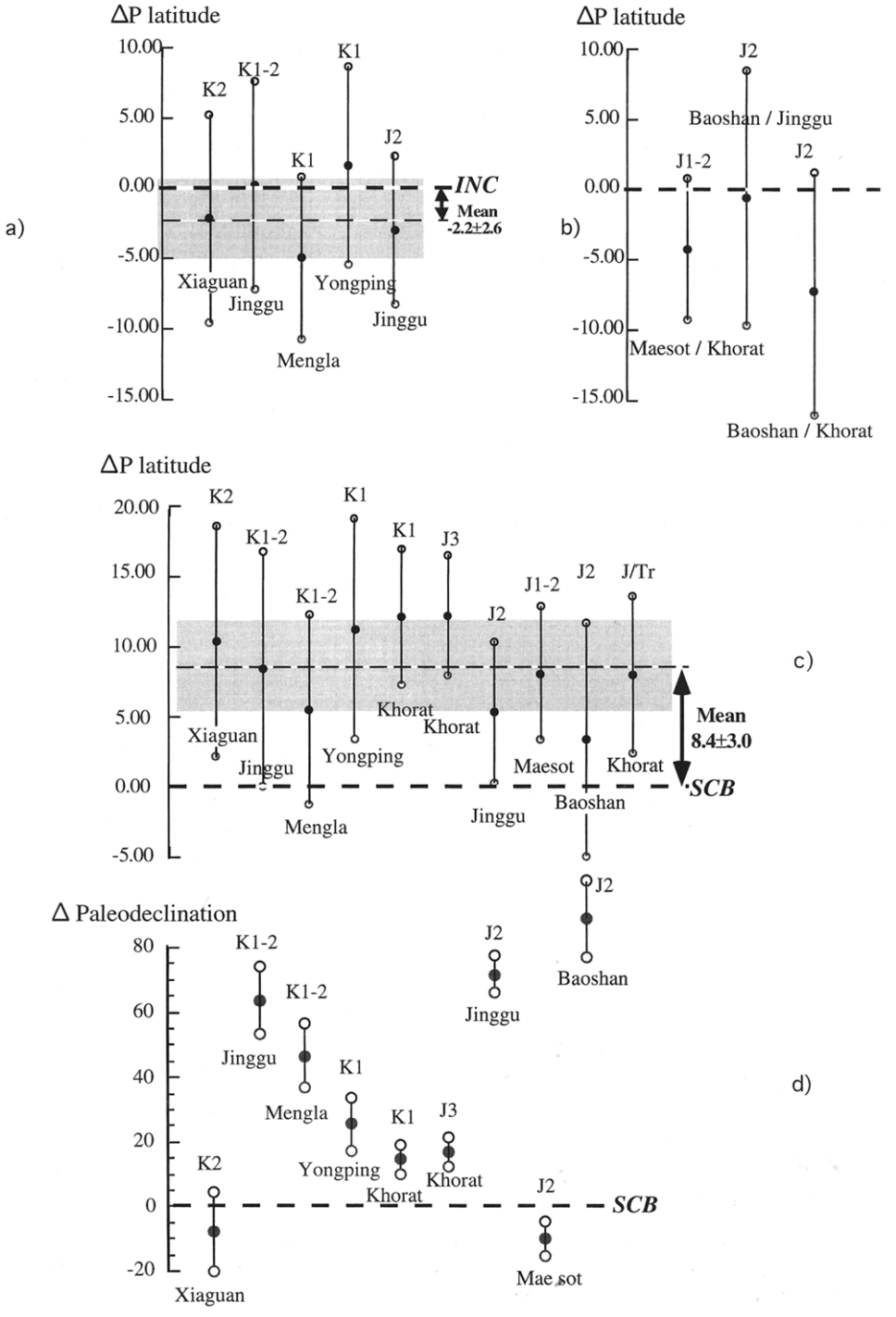
Reliable Mesozoic paleomagnetic results from the Baoshan area in the northwestern part of the STM

Table 2
Paleomagnetic results from the Indochina, Simao and Shan-Thai blocks

Block	Location (°N/°E)	Age	N	D(°)/I(°)/α_{95}	λ (°N)/φ(°E)	A_{95} (or dp/dm)
Indochina	16.5/103.0	K1	10	-	62.7/173.3	2.4
Indochina	16.5/103.0	J3	10	-	64.8/178.1	2.3
Indochina	16.5/103.0	J	18	-	60.2/176.8	4.0
Indochina	16.7°/101.8	J1/T3	13	38.8/41.9/4.8	52.9/173.3	5.2
Simao(YongPing)	25.5/99.5	K1	12	42.0/51.1/15.7	50.9/167.3	20.6
(Bivariate Mean)				40.1/49.7/7.4/16.9	54.4/172.0	
Simao (Xiaguan)	25.6/100.2	K2	9	6.9/47.7/8.6	83.6/152.7	10.0
Simao (Jinggu)	23.4/100.9	K1-2	8	79.4/43.3/9.1	18.9/170.0	8.9
Simao (Mengla)	21.6/101.4	K1-2	10	60.8/37.8/7.6	33.7/179.3	8.2
Simao (Weishan)	25.4/100.2	J3	5	7.3/25.3/10.4	76.3/250.0	10.4
Simao (Jinggu)	23.6/100.5	J2	10	83.3/36.8/5.4	14.0/173.6	3.6/6.2
Shan-Thai (Baoshan)	24.3/98.4	J2	6	99.7/35.2/11.3	-0.5/166.6	7.5/13
Shan-Thai (Mae Sot)	16.7/98.7	J1-2	12	0/31.4/5.0	89.7/57.7	3.1/5.6

N = number of sites; $D/I/\alpha_{95}$ = declination/inclination/semi-ellipse of confidence at the 95% level; $\lambda/$ = VGP latitude and longitude; A_{95} = VGP confidence interval; d_p/d_m = errors in VGP latitude and longitude

block and from the nearby Yongping, Jinggu and Simao areas on the Simao block (Table 2) have been recently published by Huang and Opdyke and Funahara et al. [17,18]. These results are all derived from studies of continental Jurassic rocks and Cretaceous redbeds.



A fold test at the 95% level [26] for the Jinggu and Mengla sites of the Mangang Formation and a fold test at the 99% level along with a reversal test at 95% for the Xiaguan Formation establishes the primary nature of the magnetization for the Cretaceous sites [17]. However, the authors of these studies questioned the age of the magnetization acquisition of the Xiaguan section. Indeed, some sites are of reversed polarity while the age of the formation is supposed to be within the Cretaceous long normal interval. According to these authors, the age of the folding in this region indicates that the age of the acquisition of magnetization is pre-Eocene, and most likely Late Cretaceous. No fold test was available for the Jurassic sites, but the authors argued that the magnetization was primary because of the internal consistencies between sites. The Weishan–Bazhulu Upper Jurassic pole has not been used given the poor confidence attributed to this pole by the authors themselves.

An Early Cretaceous direction was also reported by Funahara et al. [18] from a study of 23 sites near Yongping. The primary nature of the magnetization was ascertained from a positive fold test at the 99% level on twelve sites and by the presence of two sites of reversed polarity. The mean direction found in this study has a large A_{95} of 15.7°. This somewhat large uncertainty is mainly due to a large dispersion of the declinations, probably related to local vertical axis rotations.

Preliminary results for the Khorat plateau have been reported by several authors over the last two decades [22,27–29]. A reappraisal of these data, along with a new analysis of additional Devonian to Permian samples and one Jurassic site led Chen and Courtillot [30] to suggest that most of the Khorat region may have suffered a complete remagnetization after the Triassic and possibly after the Cretaceous or the early Cenozoic. Yang and Besse [16] therefore collected not only additional Permian and Triassic samples but also Jurassic and Lower Cretaceous material from the Khorat plateau (Table 2). A negative fold test was found for the Permian succession. A characteristic direction was isolated from the Upper Triassic–Lower Jurassic formations. Two antipodal directions of normal and reversed polarity, a positive fold test at the 99% level and remanence carried both by magnetite and hematite in different lithologies indicated a primary origin of the magnetization or at least acquisition shortly after deposition. Indeed, the structural study conducted by Bunopas et al. [22] suggests a Middle–Late Jurassic age of folding in this region. These results also suggested that the magnetization of the Permian limestone area was acquired before this (and most probably before the late Norian), and thus partly confirmed the widespread remagnetization suggested by Chen et al. [30]. No significant fold test result or clear reversals were able to ascertain a primary nature for the Upper Jurassic–Lower Cretaceous Khorat plateau pole. However, its mean direction is significantly different from that of the Triassic–Jurassic pole, and also from the Neogene pole direction of McCabe et al. [31], based on a study of volcanic rocks located immediately to the south of the sampled area. Moreover, the deposition of this succession directly above the Upper Triassic–Lower Jurassic sandstones, for which a positive fold test is available, led Yang and Besse to suggest that the magnetization of Upper Jurassic–Lower Cretaceous rocks was also of primary origin.

We have plotted (Fig. 7a) latitude anomalies (defined as the differences between observed latitude values on the Simao block and those that would have

Fig. 7. Paleolatitude differences between

(a) Simao block and Khorat (predicted latitudes for Khorat are computed at the coordinates of each locality)

(b) Mae Sot and Khorat, Baoshan and Khorat, and Baoshan and Jinggu (Shan–Thai–Malay and Simao blocks)

(c) China and Indochina blocks (predicted latitudes for China are computed at the coordinates of each locality).

(d) Rotations of the Simao and Indochina blocks relative to China. Error bars for the latitude differences at each site are computed using $\sqrt{(\delta\lambda^2_{\text{observed}} + \delta\lambda^2_{\text{predicted}})}$, where $\delta\lambda$ is the paleolatitude error for rotation differences of $\sqrt{(\delta D^2_{\text{observed}} + \delta D^2_{\text{predicted}})}$, where δD is the declination error.

been expected at these sites if the sites had been rigidly attached to the Khorat region) using the directions computed in Yang and Besse [16]. A wide dispersion of the declinations in the study by Funuhara et al. [18] leads to a fairly large uncertainty, and we have thus computed a mean paleolatitude for this study based on inclinations only using McFadden and Reid's statistics [32]. We have also used for the same study the bivariate statistical method developed by Le Goff [33] which takes into account the anisotropy of directions. Using both methods the mean inclinations were found to be nearly identical ($I = 46.1 \pm 8.6°$ using McFadden and Reid and $49.7 \pm 7.4°$ using Le Goff). There is a consistent pattern in Fig. 7a, showing that the latitude anomalies are small during the Cretaceous and Jurassic. At the 95% confidence level no significant differences can be observed. The arithmetic mean of the latitude anomalies leads to an insignificant difference of $-2.2 \pm 2.6°$ between the Simao and Khorat blocks. This is in accord with a Late Triassic (or older) amalgamation between the Simao block and the Khorat region through the Nan–Uttaradit suture, with little or no northward component of motion.

The same conclusion can be drawn from Fig. 7b, which shows the latitude anomaly between Khorat

Table 3
Mesozoic poles from the North and South China Blocks [34,36,38,40,47–58]

Site location	Age	Pole Position λ (°N)/φ(°E)/A95			Ref	Author
40.1/112.9	Ku	79.6	/170.1	5.8	47	Zheng et al. (1990)
25.0/116.4		67.9	/186.2	9.2	48	Hu et al. (1991)
32.0/119.0		76.3	/172.6	10.3	49	Kent et al. (1986)
30.0/102.9		72.8	/241.1	5.0	50	Enkin et al. (1991)
23.0/115.0		66.0	/221.5	3.0	51	Hsu (1987)
26.6/102.4*		78.9	/186.6	5.5	52	Zhu et al. (1988)
26.5/102.3*		81.9	/ 220.9	7.1	53	Huang and Opdyke (1992)
	Ku Mean		n=5			74.3 /202.4 9.9
35.0/108.0	Kl	75.8	/208.7	7.5	36	Ma et al. (1993)
29.7/120.3		77.1	/227.6	5.5	54	Lin et al. (1985)
30.0/102.9		74.5	/229.0	3.2	50	Enkin et al. (1991)
30.0/103.0		78.6	/201.0	15.3	55	Zhuang et al. (1988)
45.4/107.6		72.5	/205.3	7.6	56	Pruner (1992)
42.0/119.2		82.9	/249.5	5.7	34	Zhao et al. (1990)
26.0/117.2		66.9	/221.4	5.4	57	Zhai et al. (1992)
26.8/102.5*		69.0	/204.0	4.3	53	Huang and Opdyke (1992)
	Kl Mean		n=7			75.8 /218.4 4.6
	Ku+Kl mean		n=12			75.3 /211.4 4.3
29.4/120.0	J3	73.0	/213.7	12.6	54	Lin et al. (1985)
29.9/102.9	J3	74.6	/235.3	13.2	50	Enkin et al. (1991)
26.2/101.5*	J3	66.2	/187.6	7.6	51	Zhu et al. (1988)
28.5/105.0	J1-2	78.8	/229.2	4.0		Yang et al (in preparation)
22.0/108.7	J1	78.0	/251.8	6.3	40	Gilder et al. (1993)
22.6/108.5	J1	79.0	/225.8	14.0	40	Gilder et al. (1993)
40.1/112.9	J2	76.2	/199.9	8.3	47	Zheng et al. (1990)
38.6/112.0	J2	76.1	/214.6	13.0	58	Fang et al. (1988)
36.7/109.2	J2	74.3	/232.8	4.5	38	Yang et al. (1992)
36.2/109.3	J1	82.4	/286.0	6.8	38	Yang et al. (1992)
	J1+J2 Mean		n=7			78.8 /230.0 4.2
	Global Jurassic Mean		n=9			77.7 /228.3 3.6

and the STM block (Mae Sot and Baoshan areas). There is a 5° anomaly that may indicate northward motion sometime in the Middle–Late Jurassic between the STM and the Simao–Khorat blocks through the 'cryptic' Changning–Menglian suture. However, this motion has not yet been shown to be statistically significant. A direct comparison between Baoshan and Jinggu which, are the nearest sites on

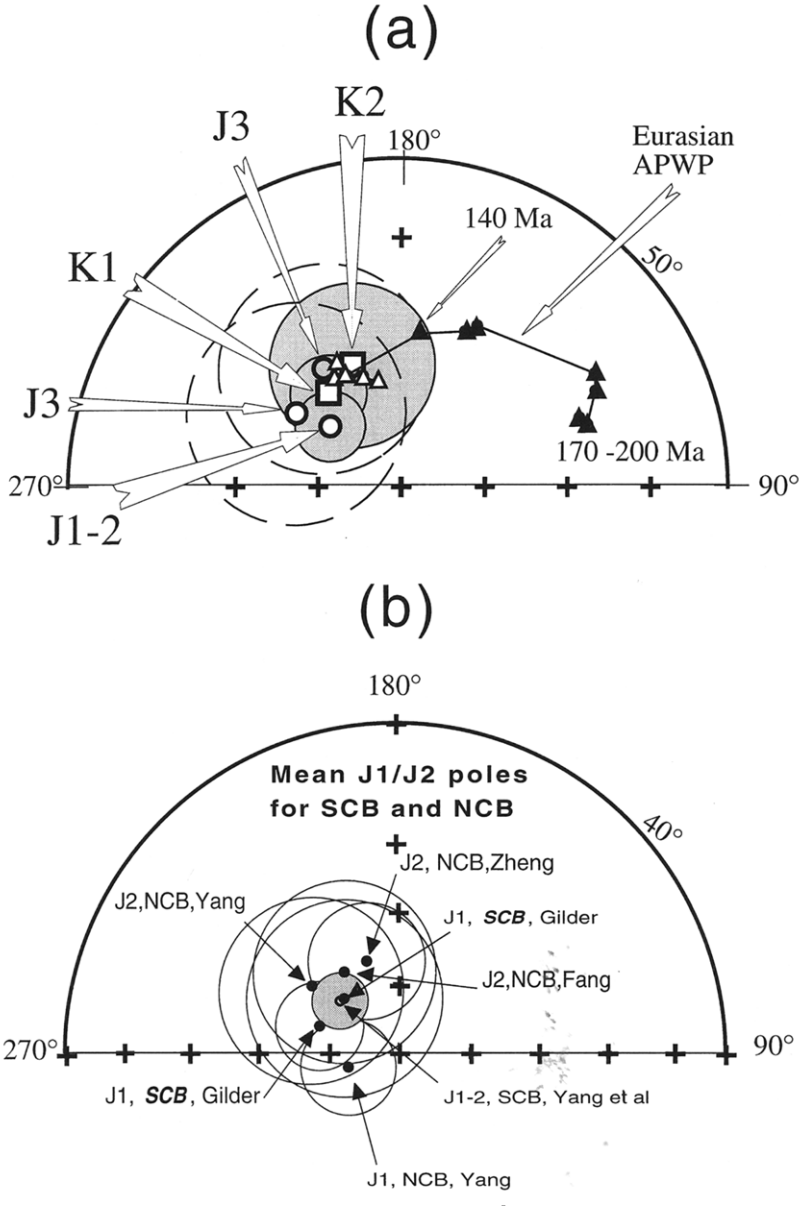


Fig. 8. Equal-area projections. (a) Late (K2) and Early (K1) Cretaceous (□) and Late Jurassic (J3) and Early–Middle (J1) Jurassic (○) paleopoles from the North and South China Blocks with their 95% half-angle of confidence. Also figured are the synthetic Eurasian paleopoles for 70–200 Ma [39]: △ = Cretaceous poles; ▲ = Jurassic poles. (b) J1 and J2 paleopoles for NCB and SCB. Mean Early–Middle Jurassic pole interval of confidence is marked in light grey.

the STM and Simao blocks respectively (Fig. 7b), led Huang and Opdyke [17] to conclude that no motion occurred between these two blocks. More data on these blocks will be necessary in order to reduce the still large uncertainties. The problem of a late accretion between the STM and the Simao–Khorat blocks remains, thus, open still.

4.2. Comparison with China

Many paleomagnetic studies in the last decade have focused on China. APWPs for the North China and South China blocks (NCB and SCB) have been extensively discussed in a large number of publications [16,34–38]. Most of the data are of good quality (at least for certain periods), and can be safely used as a reference for comparison with other blocks. The studies we have used are quoted in Table 3. The evolution of the Chinese blocks during the Jurassic and Cretaceous can be summarized as follows: During the Late and Early Cretaceous there is excellent agreement between the poles for Mongolia and the NCB and SCB [34–37]. These blocks were welded together in a position close to the that of the present. Fig. 8a shows the Chinese mean poles of Late and Early Cretaceous age which are indistinct at the 95% probability level. The two Cretaceous poles for these regions are also indistinct from the Cretaceous segment of the synthetic APWP of Besse and Courtillot [39] for Eurasia, which shows a remarkable standstill. Comparison of these data shows that the NCB, the SCB and Mongolia were part of Eurasia at that time.

New Early and Middle Jurassic poles from the SCB by Gilder et al. [40] and Yang et al. (in prep.) are in good agreement with other poles from the NCB, with the exception of Early Jurassic poles for Yunnan. This region, which is located west of the Xian–Shui–He fault system, and is known to have been significantly deformed during the India–Asia collision, has suffered important tectonic strain that resulted in noticeable rotations about vertical axes. The virtual geomagnetic poles from this area were not included, as discussed in Yang and Besse [16]. Early–Middle Jurassic data (Fig. 8b) indicate that the SCB and NCB were already welded as a single continent through the Quinling Mountain Belt at that time. A mean global Jurassic pole has been computed that also uses the two poles of Late Jurassic age, because the Early–Middle and Late Jurassic means are very close to each other (Fig. 8a). The Jurassic and Cretaceous poles from China remain in the same area, emphasizing a remarkable standstill of these regions for a large part of the Mesozoic. The Jurassic poles are remote from the Jurassic segment of the Eurasian APWP (Fig. 8a), and imply a Late Jurassic collision of the NCB/Mongolia against Siberia [35,38]. We then compare all individual paleomagnetic directions from the Khorat and Simao blocks with the coeval directions from China, computed using the mean poles for the Cretaceous and the Jurassic. In Fig. 7c we have plotted the latitude anomalies defined as the differences between observed latitude values on the Khorat and Simao blocks and those that would have been expected at these sites if the sites had been rigidly attached to China. Data from the STM block are also shown because they revealed no significant latitude anomaly relative to the Khorat region. Eight data out of ten show a paleolatitude anomaly which is significant at the 95% level. Only for the K1–2 Mengla and the J2 Baoshan studies is such a level of confidence not the case, although the mean values obtained for these studies do agree in sign and magnitude with the other means. There is no noticeable trend in the latitude anomaly between the Jurassic and Cretaceous. The overall mean difference is $8.4 \pm 3°$, corresponding to a significant southward motion of Indochina relative to South China since the Cretaceous. This value is comparable (but has a smaller uncertainty) to the values found in the Khorat paleomagnetic study of Yang and Besse [16], which are $11.5 \pm 6.7°$ for the Jurassic/Cretaceous boundary between the SCB and Indochina and $7.9 \pm 5.6°$ for the Triassic/Jurassic boundary between Huili and Indochina. By taking a paleoazimuth of 45° for the Red River Fault, this 8° southward motion translates to 1200 ± 500 km of sinistral motion on this fault, implying a large post-Cretaceous extrusion of the Indochina block. This value is much greater (although compatible in the extreme limits of the error bars) than the < 700 km determined from the kinematics of the South China Sea and the analysis of geological marker offsets proposed by the Tapponnier group [6,41,42]. Radiochronological studies of rocks from the RRF [42,43] have indicated a 32–16

m.y. age range of motion, synchronous with the age of magnetic anomalies in the China Sea [41]. A larger amount of motion would thus require an earlier phase of extrusion. This phase would be initiated immediately after the onset of the collision. Such a possibility is perhaps indicated by the 50–60 Ma ages recently found by Zhang in the north of the RRF [44], and is compatible with an early age of collision around 65–60 Ma, as proposed, for example, by Jaeger et al. [45].

4.3. Rotations

The rotations of various sites in Indochina relative to China are shown in Fig. 7d. Most rotations are clockwise, only two are counterclockwise (and small). These regions are widely deformed, and an important problem is knowing the size of the relatively undeformed crustal block that a particular paleomagnetic study represents. For the Mae Sot area, a counterclockwise rotation of $10 \pm 5.5°$ is found relative to China. This region is bounded by two major faults (Mae-Ping and the Three Pagodas) that are known to have acted successively as left and then right-lateral strike-slip faults during the Tertiary. Unmapped faults between these two zones also probably exist. Given the complicated structural setting of this region, any sense and amount of rotation is possible and we consequently feel that the small counterclockwise rotation is probably only of local significance.

Clockwise rotations of 14–73° relative to China are found for the other studies, showing that these regions did deform internally (Fig. 7d). These conclusions were also reached by Huang and Opdyke [17]. A clockwise rotation of $14 \pm 7°$ of the Khorat region relative to the SCB was found by Yang and Besse [16]. This is in good agreement with the 12° rotation found by Briais et al. [41] on the basis of interpretation of magnetic anomalies in the South China Sea. The large size of the continental blocks and the consistent rotations derived from oceanic anomaly and paleomagnetic data probably preclude the large-scale domino models of deformation proposed by Dewey et al. [46] and England and Molnar [19]. Greater clockwise rotations (between 20 and 60°) are found for sites in the Simao, Lamping and Baoshan basins. These regions were located closer to the collision zone, possibly on the easternmost front of the Indian indenter, and were subsequently more deformed and rotated (but over a small area).

5. Conclusion

We have sampled the Lower–Middle Jurassic Mae Sot Formation of the Shan–Thai–Malay (STM) block in order to test a possible Late Jurassic accretion of this block against the Simao and Khorat blocks. At the 95% level no significant latitudinal motion between these blocks is presently required by the data. Further work on the STM block will be necessary to test the hypothesis that it accreted to Simao as late as in the Late Jurassic. Comparison of paleomagnetic data from the Simao block and Indochina reveal no significant relative latitudinal motion. This is in accord with a Late Triassic (or older) amalgamation of these regions through the Nan–Uttaradit suture. Relative rotations between these regions range between 14 and 73°, and imply large, if local deformation. Comparison of these data with those from China, from Upper Triassic to Upper Cretaceous rocks, demonstrates a $8.4 \pm 3°$ southward motion of terranes situated west of the Red River Fault. The total sinistral motion on this fault is found to be of the order of 1200 km. This is in accord with a previous estimate by Yang and Besse [16] based on paleomagnetic data from the Khorat plateau only and further strengthens the extrusion model of continental deformation initiated during the India–Asia collision [6,7,41]. Such a large value of motion on the RRF would, however, imply an earlier age of extrusion than that estimated by several authors.

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